

Shaun John Thomas

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PROFILE

Final-year Aero-Mechanical Engineer with strong aerodynamic fundamentals and hands-on experience in Computational Fluid Dynamics (CFD) and stability analysis, currently applying this to the design and upcoming manufacture and flight testing of a Blended Wing Body Unmanned Aerial Vehicle (UAV). Directed a 4-member team through the design, Geometric Dimensioning and Tolerancing (GD&T)-based manufacturing and build of a 10,000 ft model rocket, coordinating sub-systems from structures to propulsion. Proficient across the full simulation-to-hardware pipeline, with proven cross-functional collaboration in multidisciplinary student engineering teams.

EDUCATION

University

BEng Aero-Mechanical Engineering, University of Strathclyde
Current Average: 69%

Glasgow, UK | September 2025- Present

BTech Mechanical Engineering, Vellore Institute of Technology (Transferred)
2 Year CGPA: 8.4/10

Chennai, India | August 2023 – June 2025

EXPERIENCE AND PROJECTS

Blended Wing Body UAV Project

Glasgow, UK | August 2026 - Present

Individual Project

- Evaluated initial BWB geometries using a Clark Y airfoil in XFLR5, identifying inherent pitch stability limitations and successfully pivoting to an MH series reflexed profile to achieve natural aerodynamic trim.
- Redesigned macro-geometric parameters in SolidWorks, iteratively adjusting wing twist, sweep angles, and chord lengths to reconcile conflicting requirements between internal payload volume and target lift/drag ratios.
- Validating the optimized MH series configuration through advanced CFD sweeps in ANSYS, targeting quantifiable reductions in induced drag and a positive static margin prior to fabrication.
- Developing a physical prototyping pipeline to transition the digital model into a flight-test article, planning to leverage a combination of 3D-printed internal structures, CNC tooling, and composite skin layups for a rigid, lightweight airframe.

Airframe Design Engineer

Glasgow, UK | September 2025 - Present

StrathAIS, Student Rocketry Team, University of Strathclyde

- Engineering and manufacturing bulkhead and supporting structures for a 30,000 ft, Mach 1+ rocket, presenting design and verification evidence through PDR and CDR reviews for IREC 2026.
- Running FEA simulations to shape critical design choice and maximize strength to weight ratio.
- Recommending and implementing changes in manufacturing decisions to minimize error and material waste with the experience gained from previous failures.

Airframe Lead

Chennai, India | August 2023 – June 2025

Team Ignition, Student Rocketry Team, Vellore Institute of Technology

- Directed a team of 4 to design and build the airframe of a model rocket to reach an altitude of 10,000 feet, collaborating closely with various departments to reach innovative design compromises across the project.
- Manufactured supporting structures using power tools and CNC machines and reinforced rocket fins and nose cones with fiberglass sheets to withstand high temperatures and harsh conditions.
- Validated design choices using a testing suite, including CFD, FEA, pressure and vibration analysis with the help of technical guidance from external manufacturers on industry standards and practices.

SKILLS & CERTIFICATIONS

Skills and Soft Skills:

- CAD: SolidWorks, OnShape
- Manufacturing: 3D-Printing, Composites, GD&T
- Technical leadership & cross-department collaboration
- Project Management & Problem Solving
- Simulations: ANSYS (CFD & FEA), XFLR5, OpenFOAM
- Machining: CNC machining, power tools
- Numerical Analysis: MATLAB & SIMULINK, Python
- Design Reviews & Technical Communication

Certifications: Certified SolidWorks Design Associate (CSWA)